

Designing a Multi-Agent Research System

Objective

The most powerful AI systems aren't single agents working in isolation—they're teams of specialized agents that can coordinate, collaborate, and build on each other's expertise to solve complex problems that no individual agent could handle alone.

Scenario

You're building an AI research assistant for a consulting firm that needs to produce comprehensive market analysis reports. The system must coordinate multiple specialized agents: a web research agent that finds relevant articles and data sources, a data analysis agent that processes quantitative information, a trend analysis agent that identifies patterns and projections, a competitive intelligence agent that analyzes competitor activities, and a synthesis agent that weaves everything together into coherent reports.

The challenge is that these agents must work together intelligently. The web research agent's findings should inform what the data analysis agent focuses on. The trend analysis should build on both web research and data analysis. The competitive intelligence should consider trends and market data. And the synthesis agent needs to understand the reasoning and connections from all other agents to create truly insightful reports.

Your firm's clients expect reports that demonstrate deep, interconnected analysis—not just collections of separate findings. The system must coordinate agents to produce collaborative intelligence that exceeds what any individual agent could achieve.

Design a sophisticated multi-agent system that coordinates specialized research agents to produce comprehensive, interconnected market analysis reports through collaborative intelligence.

Instructions for learners

Task 1: Design Collaborative State Architecture

Create a state structure that enables true collaboration between agents:

Shared Research Context: Information that all agents can build upon

Agent Contributions: Structured findings from each specialized agent

Cross-Agent Insights: Connections and relationships discovered between different agents' work

Collaborative Reasoning: Shared understanding that emerges from agent interactions

Synthesis Threads: Narrative threads that connect findings across different domains

Task 2: Define Agent Coordination Patterns

Design coordination strategies for your five specialized agents:

→ Web Research Agent: Finds articles, reports, and online sources

→ Data Analysis Agent: Processes quantitative data and statistics

→ Trend Analysis Agent: Identifies patterns and future projections

→ Competitive Intelligence Agent: Analyzes competitor activities and market positioning

→ Synthesis Agent: Weaves findings into coherent, insightful reports

For each node, define:

- What information it needs from other agents to work effectively
- How it contributes to the shared understanding
- What triggers or events it responds to from other agents
- How it signals important findings to other agents

Task 3: Create Dynamic Workflow Logic

Design the coordination logic that determines how agents work together based on the research question and emerging findings:

How does the system decide which agents to activate for different types of research questions?

When should agents work in parallel vs. sequence?

How do agents trigger additional investigation by other agents?

What conditions determine when the synthesis agent should begin its work?

Create a decision framework that adapts the collaboration pattern based on the complexity and requirements of each research request.

Submission Guidelines

Submit a comprehensive design document (PDF, Word, or well-structured text) containing:

→ Collaborative State Schema: Complete state structure with detailed field descriptions

→ Agent Coordination Matrix: How each agent interacts with and depends on others

→ Dynamic Workflow Logic: Decision framework for adaptive collaboration patterns

→ Collaboration Rationale: Explanation of how your design creates emergent intelligence

Completion Criteria

State architecture that enables true collaborative intelligence, not just data sharing

Clear coordination patterns that show how agents build on each other's work

Dynamic workflow logic that adapts collaboration based on research complexity

Compelling explanation of how the system creates insights beyond individual agent capabilities

Engage in reflection: Transition to Dialogue

After submitting your multi-agent research system design, you'll engage in a structured conversation to evaluate your coordination strategies and explore how they enable collaborative intelligence in complex research scenarios.